[Total No. of Question -04] [Total No. of Pages - 02]

G H Raisoni International Skill Tech University, Pune

S.Y. B.Tech

TERM-III (Regular)

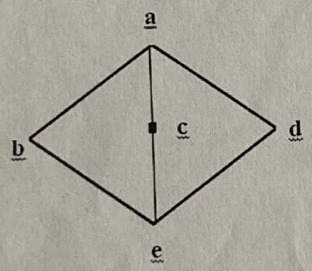
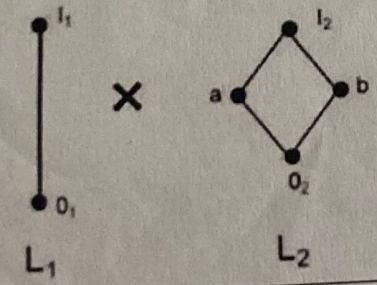
CAE II- WINTER 2025 (2024 Pattern)

DISCRETE MATHEMATICS AND GRAPH THEORY (25BTBSL301)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) All questions are compulsory. ii) neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	Question	Marks	CO	BL
Q1.(a)	Prove that (**R**,+, *) is Ring.	[5]	1	3
b)	Show that set of all divisors of 70 for divisibility relation forms a lattice. Also find maximal element and minimal element of lattice.	[5]	1	3
	OR			
Q 2. (a)	The following is the Hasse diagram of the POSET $\{\{a, b, c, d, e\}, <\}$. Is it a Lattice? Justify? (Find Composition table) 	[5]	1	3
(b)	Draw direct product of Lattice L_1 *and* L_2. 	[5]	1	3
Q 3. (a)	Find the in-degree and out-degree of each vertex in the graph G. Also find Adjacency matrix of G.	[5]	2	